FEDERAL SPECIFICATION

STARTER, FLUORESCENT LAMP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers starters for use with pre-heat type fluorescent and similar discharge lamps.

1.2 Classification.

1.2.1 Types and classes. Starters covered by this specification shall be of the following types and classes in accordance with American National Standard C78.180-1972 (see table I):

Types:

I - Glow switch with lockout.

Classes:

A - Manual reset.
B - Automatic reset.

II - Thermal switch with lockout.

Classes:

A - Manual reset.
B - Automatic reset.

III - Simple glow switch without lockout.

IV - Simple thermal switch without lockout.

1.3 Designation. Lamp application (voltage and wattage rating) and number of pins shall be as indicated in table I (see 6.1).

TABLE I. Designation of starters

Lamp application	Simple switch (without lockout)		Lockout-manual reset (Class A)		Lockout-automatic reset (Class B)	
	2 Pin	4 Pin	2 Pin	4 Pin	2 Pin	4 Pin
118-120 V Open circuit	Glow switch starters (types I and III)					
4,6,8 W	FS-5	-	-	-	-	-
14, 15 W	FS-2	-	-	-	-	-
20 W	FS-2	-	FS-20	-	FS-2NA	-
20 W	-	-	COP-20	-	FS-NB2	-
20 W	-	-	-	-	FS-20A	-
22 W circular	FS-22	-	-	-	-	-
25 W	FS-25	-	-	-	-	-
	Thermal starters (types II and IV)					
8 W	AT-58	-	-	-	-	-
20 W	AT-2	-	-	-	TC-2	-
20 W	-	-	-	-	TGC-2	-

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TABLE I. Designation of starters (Cont'd.)

Lamp application	Simple switch (without lockout)		Lockout-manual reset (Class A)		Lockout-automatic reset (Class B)	
	2 Pin	4 Pin	2 Pin	4 Pin	2 Pin	4 Pin
150 V Open circuit	Glow switch starters (types I and III)					
32 W circular	FS-12	-	-	-	-	-
85, 90, 100 W	FS-6	FS-64	COP-6	COP-64	FS-6NA	FS-64-NA
85, 90, 100 W	FS-85	FS-85-4	FS-852	FS-850	FS-85NA	FS-85NA-4
85, 90, 100 W	-	-	FS-850-2	FS-850-4	FS-850-A	FS-850A-4
85, 90, 100 W	-	-	-	FS-850S	-	-
	Thermal starters (types II and IV)					
85, 90, 100 W	-	FS-64	-	-	-	-
200 V Open circuit	Glow switch starters (types I and III)					
30 W	FS-4	-	FS-30	-	-	-
30 W	-	-	COP-30	-	-	-
40 W	FS-4	-	FS-40	-	FS-4NA	-
40 W	-	-	FS-400	-	FS-NB4	-
40 W	-	-	FS-40/400	-	-	-
40 W	-	-	COP-40	-	FS-40A	-
40 W	-	-	COP-400	-	FS-4/400/NA	-
40 W	-	-	COP-40/400	-	FS-4/400A	-
	Thermal starters (types II and IV)					
40 W	-	FS-44	TC-40	-	TC-4	-
40 W	-	-	-	-	TGC-4	-

Literal Codes in table I are to be interpreted as follows:

FS - Fluorescent Starter.

TGC - Thermal Glow Switch (Cutout type or circuit).

TC - Thermal Cutout (or circuit).

AT - Air Thermal.

COP - Cutout Protection (manual).

NA or A - Automatic.

NB - No Blink.

S - Selected (for specified characteristics).

Numeric codes in table I are, in most instances, manufacturer's codes and cannot be readily interpreted.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitations for bids, or request for proposal, form a part of this specification to the extent specified herein:

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Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fireboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes, Paperboard, Metal-Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup..

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendant of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Freight Traffic Association, Inc., Agent, 1616 P Street, NW., Washington, DC 20036.)

American National Standards Institute, Inc., Specification:

C78.180-1972 - Specifications for Fluorescent Lamp Starters.

(Copies of American National Standards Institute publications may be obtained from American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

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Underwriters' Laboratories, Inc. Standards:

UL 542 - Lampholders, Starters and Starter Holders for Fluorescent Lamps.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Qualification. Fluorescent lamp starters furnished under this specification shall be a product which has passed the qualification inspections and tests cited in 4.3, and has been listed or approved for listing on the applicable Qualified Products List.

3.2 Fire and casualty hazards.

Each supplier shall submit to the contracting agency proof that the fluorescent lamp starters he proposes to supply under this specification conform to applicable requirements of the Underwriters' Laboratories, Inc. UL 542 Lampholders, Starters, and Starter Holders for Fluorescent Lamps. The label, or listing mark of the UL may be accepted as evidence that the fluorescent lamp starter conforms to this requirement. In lieu of the label, the supplier may submit independent proof, satisfactory to the contracting agency that starters conform to applicable requirements of UL 542, including test requirements cited therein.

Compliance with the requirements of UL 542 in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the other requirements of this specification in order to secure acceptance of his product.

3.3 Material. All material used in the manufacture of the fluorescent lamp starters, shall be suitable for the purpose intended and shall be free of defects that affect the starters appearance or serviceability. Metal components shall be protected against corrosion.

3.4 Identification marking. Each starter shall be permanently and legibly marked with the manufacturer's name of trademark and catalog number or equivalent, and date of manufacture (see 4.4.4.1).

3.5 Performance and design requirements.

3.5.1 Performance. Fluorescent lamp starters covered by this specification shall be in compliance with applicable performance requirements cited in the American National Standards Institute (ANSI) Standard C78.180-1972. Testing to determine compliance shall be as prescribed in 4.3.2.2 and 4.4.4.2.

3.5.2 Radio interference suppression. In order to assist in suppression of radio interference, starters shall contain a capacitor in accordance with section 10.6 of above referenced ANSI Standard (see 4.3.2.2 and 4.4.4.2).

3.5.3 Enclosure. Unless otherwise specified (see 6.1), Starter enclosure dimensions, for each enclosure size, shall be as indicated in Figure 5 and Table 2 of ANSI C78.180-1972. Enclosure size (either medium or mogul), number of pins, and pin spacing (see Figure 5 and Table 2 of above ANSI Standard) shall be specified by user (see 6.1).

3.6 Design and workmanship. Design and workmanship shall be such that starters are free from any defects that affect appearance or serviceability or any defects that may cause injury to user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.1), the supplier shall be responsible for the performance of all inspections and tests necessary to determine compliance with this specification. The Government reserves the right to perform any inspection or test set forth herein in order to determine specification compliance.

4.2 Classification of inspections and tests.

The inspection and testing of fluorescent lamp starters covered by this specification shall be classified as follows:

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A. Qualification inspections and tests (see 4.3).

B. Lot acceptance inspections and tests (see 4.4).

4.3 Qualification inspections and tests.

Qualification inspections and tests shall include inspections and tests cited in 4.3.2. All qualification inspections and tests shall be conducted by an independent test laboratory acceptable to the Government.

4.3.1 Sampling. Qualification sample shall consist of fifty (50) starters of the same type, class, lamp application, size, and number of pins (see 1.2, 1.3 and 3.5.3). Samples shall be selected at random from the manufacturer's current production.

4.3.1.1 Acceptance Criteria. Acceptance criteria for qualification inspections and tests shall be as cited below in each respective inspection and test in 4.3.2.

4.3.2 Inspections and tests.

4.3.2.1 Visual inspections. Visual inspections as outlined below shall be performed to determine compliance with requirements for enclosure dimensions (see 3.5.3), and design and workmanship (see 3.6).

4.3.2.1.1 Inspection procedure. The fifty (50) sample starters, selected in accordance with 4.3.1 shall be visually examined to determine compliance with requirements discussed in 4.3.2.1. Failure of more than three starters to meet enclosure dimension requirements (see 3.5.3) or indication of defects affecting appearance or serviceability (see 3.6) on more than three starters shall preclude qualification. Also, indication of any defect(s) that may cause injury to the user (see 3.6) shall preclude qualification.

4.3.2.2 Performance tests. Testing to determine compliance with 3.5.1 shall be performed utilizing the test methods and equipment cited in ANSI Standard C78.180-1972.

4.3.2.2.1 Test procedures. The entire sample of fifty (50) starters, selected in accordance with 4.3.1, shall be subjected to all applicable tests cited in Sections 10.2.1, 10.2.3, 10.8, 10.9 and 10.10 of the above ANSI Standard. Subsequently, ten (10) of the fifty (50) starters shall be subjected to all of the remaining tests in Section 10 of the ANSI Standard. Type II and IV (Thermal Switch) starters shall meet the performance requirements of Sections 10.2 and 10.4 on both 60 Hertz alternating current (AC) as well as on direct current (DC), open circuit voltages being as cited in table I (see 6.1 b of this specification and Section 13 of the ANSI Standard). Failure of more than one starter to meet the requirements of Section 10.6 of ANSI Standard (see 3.5.2) shall preclude qualification. Acceptance criteria for other tests cited herein shall be as stated in Section 14 of the above ANSI Standard, C78.180-1972. Test failure, other than as permitted in Section 14, shall preclude qualification.

4.3.3 Disqualification. The Government reserves the right to remove from the Qualified Products List (QPL) (see 3.1), any starter(s) that has passed qualification inspections and tests and subsequently fails to pass lot acceptance inspections and tests (see 4.4).

4.4 Lot acceptance. Lot acceptance inspections and tests shall include all inspections and tests cited in 4.4.4.

4.4.1 Definition of lot. A lot shall consist of starters of the same type, class, lamp application, size, and number of pins, (see 1.2, 1.3, and 3.5.3).

4.4.2 Sampling for lot acceptance. Sampling for lot acceptance visual inspection (4.4.4.1) shall be inspection level II, in accordance with MIL-STD-105. Sampling for lot acceptance tests (4.4.4.2) shall be inspection level S-4, in accordance with MIL-STD-105. Samples shall be selected at random from manufacturer's current production.

4.4.3 Acceptance criteria. Acceptance criteria for inspections and tests shall be as cited for each respective inspection and test in 4.4.4.

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4.4.4 Inspections and Tests.

4.4.4.1 Visual inspection. Samples selected in accordance with 4.4.2 shall be visually inspected to determine compliance with marking (3.4), enclosure dimension requirements (3.5.3), and design and workmanship requirements (3.6). Acceptance criteria for visual inspection to determine compliance with dimension requirements (3.5.3), or to determine defects affecting appearance or serviceability (3.6) shall be in accordance with an AQL of 2.5 percent defective, as cited in MIL-STD-105. Indication of any defect that may cause injury to user (3.6) shall be cause for lot rejection. Any starter failing to meet any visual inspection requirement as cited above shall be considered defective.

4.4.4.2 Tests. Testing to determine compliance with 3.5.1 shall be performed utilizing the test methods and equipment cited in ANSI Standard C78.180-1972.

4.4.4.2.1 Test procedure. All sample starters selected in accordance with 4.4.2, shall be subjected to all applicable tests cited in Sections 10.2.1, 10.2.3, 10.9 and 10.10 of the above ANSI Standard. Subsequently, twenty (20) percent of the sample starters shall be subjected to all remaining tests in Section 10 of the ANSI Standard, with exception of test cited in Section 10.8. Type II and IV (Thermal Switch) starters shall meet the performance requirements of Section 10.2 and 10.4 on both 60 Hertz alternating current (AC) as well as on direct current (DC), open circuit voltages being as cited in table I (see 6.1 b of this specification and Section 13 of the ANSI Standard). Failure of more than ten (10) percent of starters tested to determine compliance with requirements of Section 10.6 of ANSI Standard shall be cause for lot rejection. Acceptance criteria for other tests cited herein shall be as stated in Section 14 of the above ANSI Standard C78.180-1972. Test failure other than as permitted in Section 14 shall be cause for lot rejection.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with Table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 per hundred units.

Table II

Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Ommited, incorrect, illegible; improper size, location, sequence or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps. Distortion of containers.
Container	Not as specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Ten starters of like description shall be packaged in accordance with method 1C-3 of MIL-P-116.

5.1.1.2 Intermediate packaging. Ten unit packages, as specified in 5.1.1.1 shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

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5.1.2 Level B.

5.1.2.1 Unit packaging. Ten starters of like description shall be packaged in a box conforming to PPP-B-566, PPP-B-636, PPP-B-665, or to PPP-B-676. Each box shall be secured to prevent accidental opening during shipment and storage.

5.1.2.2 Intermediate packaging. Ten unit packages, as specified in 5.1.2.1 shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with the appendix to PPP-B-636.

5.1.3 Level C.

5.1.3.1 Unit packaging. The starters shall be packaged to insure protection during shipment and safe delivery to destination.

5.1.3.2 Intermediate packaging. The starters packaged as specified in 5.1.3.1 shall be further packaged in multiples to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Starters of like description, packaged as specified in 5.1.1, shall be packed in quantities as specified in the contract or order in a box conforming to PPP-B-636, class weather resistant, grade DW or to PPP-B-640, class 2. Close, waterproof seal and strap the box in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Ten intermediate packages of starters of like description (1000 starters), packaged as specified in 5.1.2, shall be packed in a box conforming to PPP-B-636, class domestic, grade 275 minimum. Closure of the box shall be in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. Starters, packaged as specified in 5.1.3, shall be packed to comply with the Uniform Freight Classification Rules or the National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.1).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Type and class (see 1.2.1). Types II and IV starters should be used for locations in which the temperature may be low, where the voltage may vary excessively, and for direct current systems.
(c) Lamp application (voltage and wattage rating), number of pins, and whether Medium Size or Mogul Size (see 1.2.1, 1.3 and 3.5.3).
(d) Starter dimensions, if other than specified (see 3.5.3).
(e) Responsibility for inspection, if other than specified (see 4.1).
(f) Level of packaging and packing required (see 5.1 and 5.2).
(g) Marking required (see 5.3).

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required test and found satisfactory for inclusion on the Federal Qualified Products List. The attention of supplier is called to this requirement, and such suppliers are urged to communicate with the Standardization Division, Federal Supply Service, General Services Administration, Washington, DC 20406, and arrange to have the products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Standardization Division, Federal Supply Service, General Services Administration, Washington, DC 20406.

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MILITARY CUSTODIANS:

Army - ME
Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:

Commerce - NBS
DC/Gov't - DCG

Review Activities:

Army - ME

User Activities:

MC, YD-H

PREPARING ACTIVITY: GSA - FSS

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